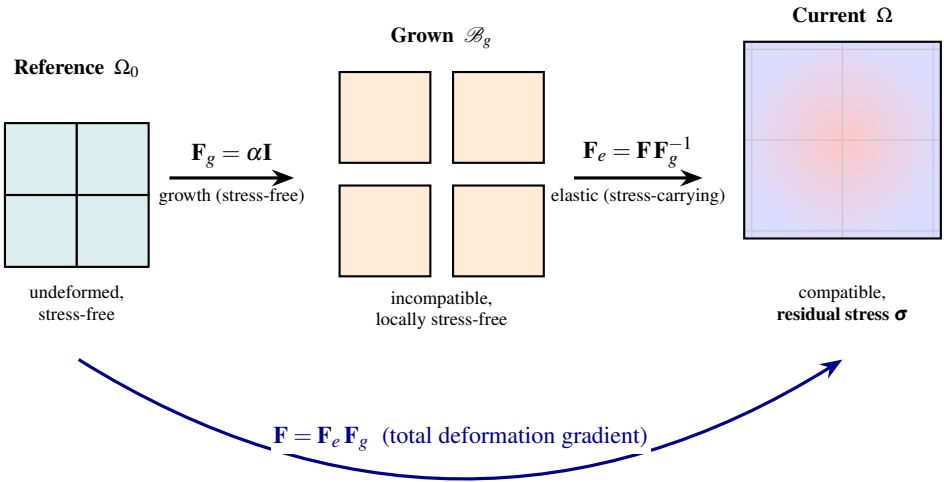


Multiplicative split (Rodriguez et al. 1994; Klempt 2024): isotropic growth $\mathbf{F}_g = \alpha\mathbf{I}$ with $\alpha(0) = 1$, $\det\mathbf{F}_g = \alpha^3$ (volume added); the intermediate $\mathscr{B}_g$ is a locally relaxed, *incompatible* state. Compatibility is restored by the elastic part $\mathbf{F}_e$, which alone carries stress ($\boldsymbol{\sigma} = J_e^{-1}\,\partial_{\mathbf{F}_e}\psi\,\mathbf{F}_e^{\top}$, Neo-Hookean). Under geometric constraint (substrate adhesion / confinement) the growth incompatibility yields growth-induced residual stress, which the material routine returns.